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Financial Analysis Techniques

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PROFITABILITY RATIOS

Learning Outcome

- calculate and interpret activity, liquidity, solvency, and profitability ratios

The ability to generate profit on capital invested is a key determinant of a company's overall value and the value of the securities it issues. Consequently, many equity analysts consider profitability to be a key focus of their analytical efforts.

Profitability reflects a company's competitive position in the market, and by extension, the quality of its management. The income statement reveals the sources of earnings and the components of revenue and expenses. Earnings can be distributed to shareholders or reinvested in the company.

Calculation of Profitability Ratios

Profitability ratios measure the return earned by the company during a period. Exhibit 27 provides the definitions of a selection of commonly used profitability ratios, some of which were introduced in earlier modules in Corporate Issuers. Return-on-sales profitability ratios express various subtotals on the income statement (e.g., gross profit, operating profit, net profit) as a percentage of revenue. Essentially, these ratios constitute part of a common-size income statement discussed earlier. Return on investment profitability ratios measure income relative to assets, equity, or total capital employed by the company. For operating ROA, returns are measured as operating income, that is, prior to deducting interest on debt capital. For ROA and ROE, returns are measured as net income, that is, after deducting interest paid on debt capital. For return on common equity, returns are measured as net income minus preferred dividends (because preferred dividends are a return to preferred equity).

Exhibit 27: Definitions of Commonly Used Profitability Ratios

Profitability Ratios	Numerator	Denominator
Return on Sales[a]		
Gross profit margin	Gross profit	Revenue
Operating profit margin	Operating income[b]	Revenue
Pretax margin	EBT (earnings before tax but after interest)	Revenue
Net profit margin	Net income	Revenue
Return on Investment		
Operating ROA	Operating income	Average total assets
ROA	Net income	Average total assets
Return on invested capital	EBIT × (1 - Effective Tax Rate)	Average total short- and long-term debt and equity
ROE	Net income	Average total equity
Return on common equity	Net income – Preferred dividends	Average common equity

[a] "Sales" is being used as a synonym for "revenue."

[b] Some analysts use EBIT as a shortcut representation of operating income. Note that EBIT, strictly speaking, includes non-operating items such as dividends received and gains and losses on investment securities. Of utmost importance is that the analyst compute ratios consistently whether comparing different companies or analyzing one company over time."

Interpretation of Profitability Ratios

Gross profit margin indicates the percentage of revenue available to cover operating and other expenses and to generate profit. Higher gross profit margin indicates some combination of higher product pricing and lower product costs. The ability to charge a higher price is constrained by competition, so gross profits are affected by (and usually inversely related to) competition. If a product has a competitive advantage (e.g., superior branding, better quality, or exclusive technology), the company is better able to charge more for it. On the cost side, higher gross profit margin can also indicate that a company has a competitive advantage in product costs.

Operating Profit Margin

Operating profit is calculated as gross profit minus operating costs. So, an **operating profit margin** increasing faster than the gross profit margin can indicate improvements in controlling operating costs, such as administrative overheads. In contrast, a declining operating profit margin could be an indicator of deteriorating control over operating costs.

Pretax Margin

Pretax income (also called "earnings before tax" or EBT) is calculated as operating profit minus interest, and the **pretax margin** is the ratio of pretax income to revenue. The pretax margin reflects the effects on profitability of leverage and other (non-operating) income and expenses. If a company's pretax margin is increasing primarily as a result of increasing amounts of non-operating income, the analyst should evaluate whether this increase reflects a deliberate change in a company's business focus and, therefore, the likelihood that the increase will continue.

Net Profit Margin

Net profit, or net income, is calculated as revenue minus all expenses. Net income includes both recurring and non-recurring components. Generally, the net income used in calculating the net profit margin is adjusted for non-recurring items to offer a better view of a company's potential future profitability.

ROA

ROA measures the return earned by a company on its assets. The higher the ratio, the more income is generated by a given level of assets. Most databases compute this ratio as follows:

$$\frac{\text{Net income}}{\text{Average total assets}}.$$

An issue with this computation is that net income is the return to equity holders, whereas assets are financed by both equity holders and creditors. Interest expense (the return to creditors) has already been subtracted in the numerator. Some analysts, therefore, prefer to add back interest expense in the numerator. In such cases, interest must be adjusted for income taxes because net income is determined after taxes. With this adjustment, the ratio would be computed as follows:

$$\frac{\text{Net income} + \text{Interest expense}(1 - \text{Tax rate})}{\text{Average total assets}}.$$

Alternatively, some analysts elect to compute ROA on a pre-interest and pre-tax basis (operating ROA in Exhibit 27) as follows:

$$\frac{\text{Operating income or EBIT}}{\text{Average total assets}}.$$

In this ROA calculation, returns are measured prior to deducting interest on debt capital (i.e., as operating income or EBIT). This measure reflects the return on all assets invested in the company, whether financed with liabilities, debt, or equity. Whichever form of ROA is chosen, the analyst must use it consistently in comparisons to other companies or time periods.

operating ROA, returns are measured prior to deducting interest on debt capital (i.e., as operating income or EBIT).

ROE

ROE measures the return earned by a company on its equity capital, including minority equity, preferred equity, and common equity. As noted, return is measured as net income (i.e., interest on debt capital is not included in the return on equity capital). A variation of ROE is return on common equity, which measures the return earned by a company only on its common equity.

Both ROA and ROE are important measures of profitability and will be explored in more detail later. As with other ratios, profitability ratios should be evaluated individually and as a group to gain an understanding of what is driving profitability (operating versus non-operating activities). Example 12 demonstrates the evaluation of profitability ratios and the use of the management report (sometimes called management discussion and analysis or management commentary) that accompanies financial statements to explain the trend in ratios.

EXAMPLE 12

Evaluation of Profitability Ratios

Recall from Example 1 that an analysis found that Apple's gross margin declined over the three-year period FY2015 to FY2017. An analyst would like to further explore Apple's profitability using a five-year period. He gathers the following revenue data and calculates the following profitability ratios from information in Apple's annual reports, as shown in Exhibit 28:

Exhibit 28: Profitability Ratios for Apple (US dollar millions)

	2017	2016	2015	2014	2013
Sales	229,234	215,639	233,715	182,795	170,910
Gross profit	88,186	84,263	93,626	70,537	64,304
Operating income	61,344	60,024	71,230	52,503	48,999
Pre-tax income	64,089	61,372	72,515	53,483	50,155
Net income	48,351	45,687	53,394	39,510	37,037
Gross profit margin	38.47%	39.08%	40.06%	38.59%	37.62%
Operating income margin	26.76%	27.84%	30.48%	28.72%	28.67%
Pre-tax margin	27.96%	28.46%	31.03%	29.26%	29.35%
Net profit margin	21.09%	21.19%	22.85%	21.61%	21.67%

1. Evaluate the overall trend in Apple's profitability ratios for the five-year period.

Hide Solution

Solution:

Sales had increased steadily through 2015, dropped in 2016, and rebounded somewhat in 2017. As noted in Example 1, the sales decline in 2016 was related to a decline in iPhone sales and weakness in foreign currencies. Margins also rose from 2013 to 2015 and declined in 2016. However, in spite of the increase in sales in 2017, all margins declined slightly indicating costs

Question Set

1. Which ratio would a company *most likely* use to measure its ability to meet short-term obligations?

   A. Current ratio
   B. Payables turnover
   C. Gross profit margin

   Hide Solution

   Solution:

   A is correct. The current ratio is a liquidity ratio. It compares the net amount of current assets expected to be converted into cash within the year with liabilities falling due in the same period. A current ratio of 1.0 would indicate that the company would have just enough current assets to pay current liabilities.

2. Which of the following ratios would be *most* useful in determining a company's ability to cover its lease and interest payments?

   A. ROA
   B. Total asset turnover
   C. Fixed charge coverage

   Hide Solution

   Solution:

   C is correct. The fixed charge coverage ratio is a coverage ratio that relates known fixed charges or obligations to a measure of operating profit or cash flow generated by the company. Coverage ratios, a category of solvency ratios, measure the ability of a company to cover its payments related to debt and leases.

3. Assuming no changes in other variables, which of the following would decrease ROA?

   A. An increase in average assets
   B. A decrease in interest expense
   C. A decrease in the effective tax rate

   Hide Solution

   Solution:

   A is correct. Assuming no changes in other variables, an increase in average assets (an increase in the denominator) would decrease ROA. A decrease in either the effective tax rate or interest expense, assuming no changes in other variables, would increase ROA.

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